

Narrow-line AGN selection in quiescent galaxies: exploring AGN feedback in the early universe

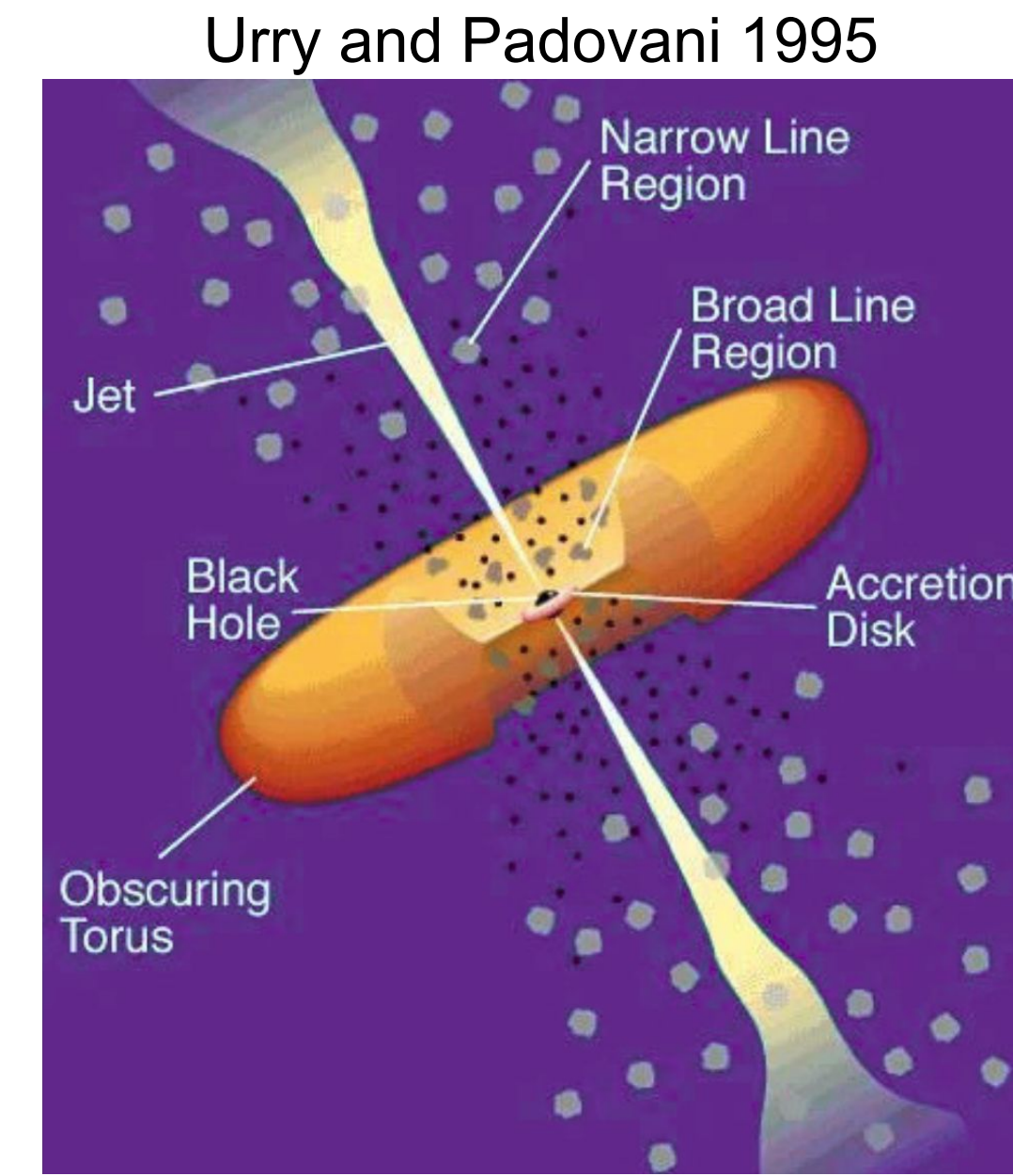
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Introduction

The formation mechanism of **quiescent galaxies**, or galaxies with little to no star formation, in the early universe remains unclear [1]. The primary proposed mechanism shutting down their star formation is feedback from **active galactic nuclei** (AGN), which are black holes in their accretion phase. The relationship of these AGN and their host galaxies in the early universe remains uncertain, which is especially true for **narrow-line AGN** (NLAGN), which are AGN observed at an angle where the dusty torus obscures the accretion disk. These objects are underexplored in JWST data, specifically in the early universe [2].



Research Goal

We aim to investigate the quenching of early galaxies by AGN feedback through studying the relationship of NLAGN and their host quiescent galaxies. We present a sample of **958 candidate quiescent galaxies** within the first few billion years of the universe from James Webb Space Telescope (JWST) Red Unknowns: Bright Infrared Extragalactic Survey (RUBIES) [3] spectroscopy and Uniform Near-Infrared Catalogs from Robust imagiNg (UNICORN) photometry. We calculate the galaxies' properties and **star formation rates** (SFRs) using Bayesian methods and seek to explore their quenching timescales and search for evidence of AGN feedback.

Methods

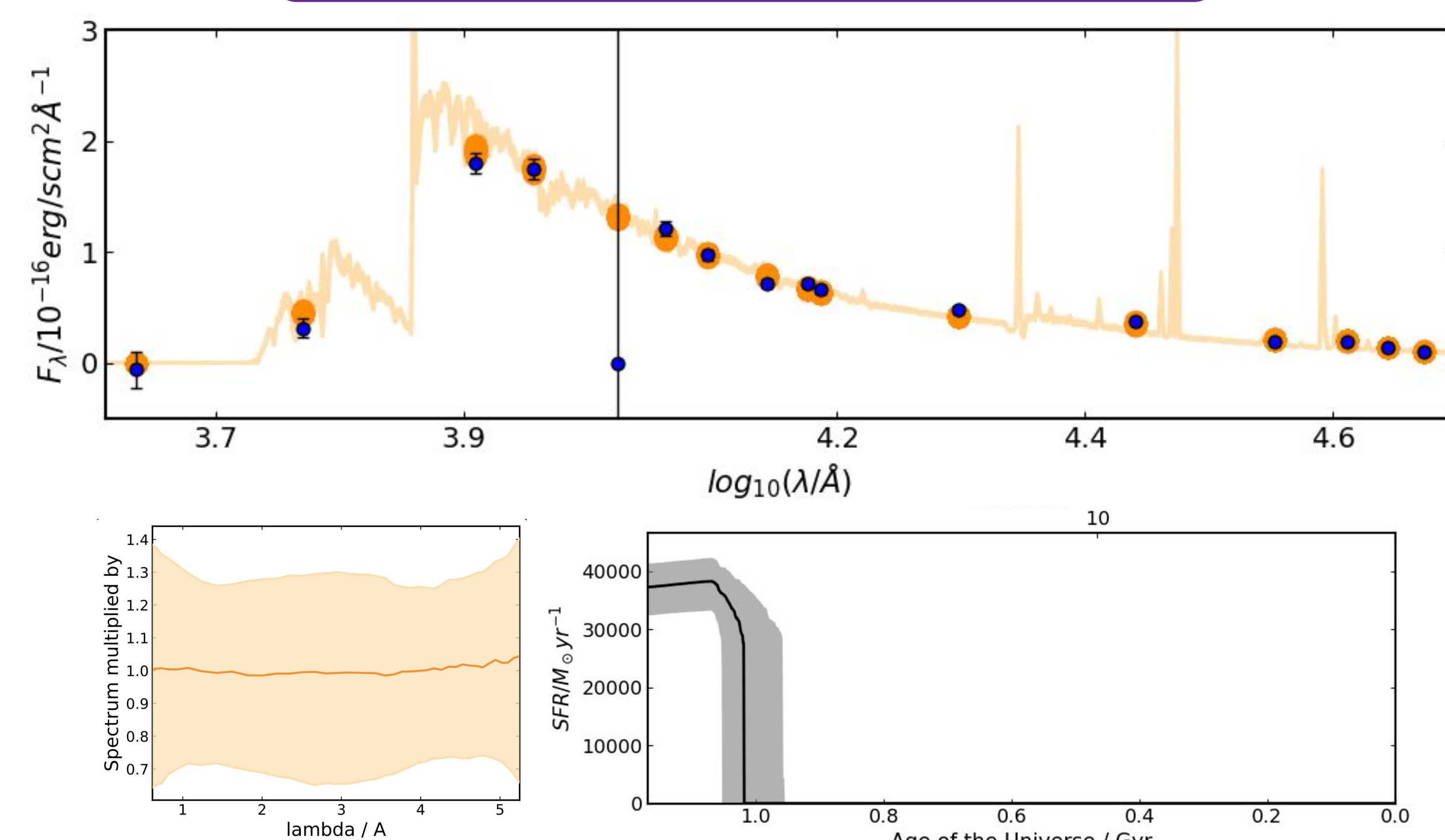


Figure 1. BAGPIPES SED fitting diagnostics. a) Successful model spectra fit to spectrophotometric data. b) Spectrum to photometry ratio. Spectrum multiplied by values around 1 imply that the spectroscopic and photometric data match. c) SFR versus cosmic time. This galaxy completely stopped star formation around 1 billion years ago.

To retrieve the galaxy properties and SFR, we use **BAGPIPES** [4], Bayesian Analysis of Galaxies for Physical Inference and Parameter Estimation, which fits model galaxy spectra to the spectrophotometric data. We assume an **exponentially declining SFR** and allow the metallicity, nebular emission, dust magnitude, and velocity dispersion to vary. We use the returned specific SFR (sSFR), which is normalized using stellar mass, to determine if galaxies are truly quiescent.

Figures and Results

We first use the **rest-frame colors** to select quiescent galaxy candidates. U, V, and J magnitudes are calculated with Lazy [5], a photometric redshift fitting code, using Markov Chain Monte Carlo (MCMC) sampling of the photometric flux. Using the Baker et al 2025 **UVJ pre-selection criteria** [6], we identify 958 candidate quiescent galaxies from $1 < z < 9$.

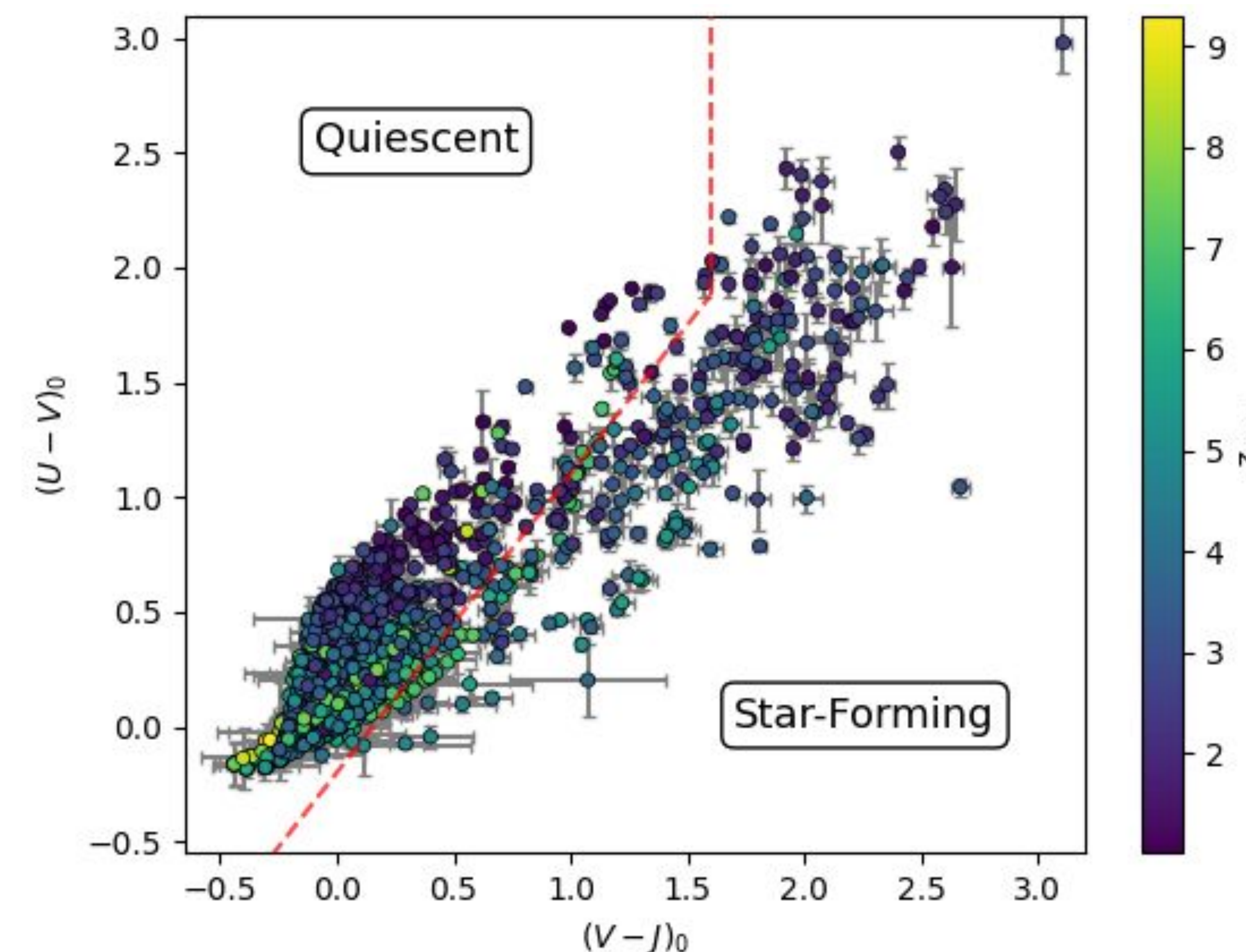


Figure 2. Rest-frame UV versus rest-frame VJ colors. Points are colored by spectroscopic redshift. The red dashed line represents the UVJ pre-selection criteria from Baker et al 2025. Sources in the top-left corner are classified as a quiescent candidate. We find 958 out of 1244 sources to be considered quiescent candidates.

From the sSFR, we find that all candidate galaxies are truly quiescent. The returned galaxy properties from BAGPIPES reveal that our sample consists primarily of **massive, dust-free galaxies**. The distribution of velocity dispersions indicates that this sample may not contain AGN outflows.

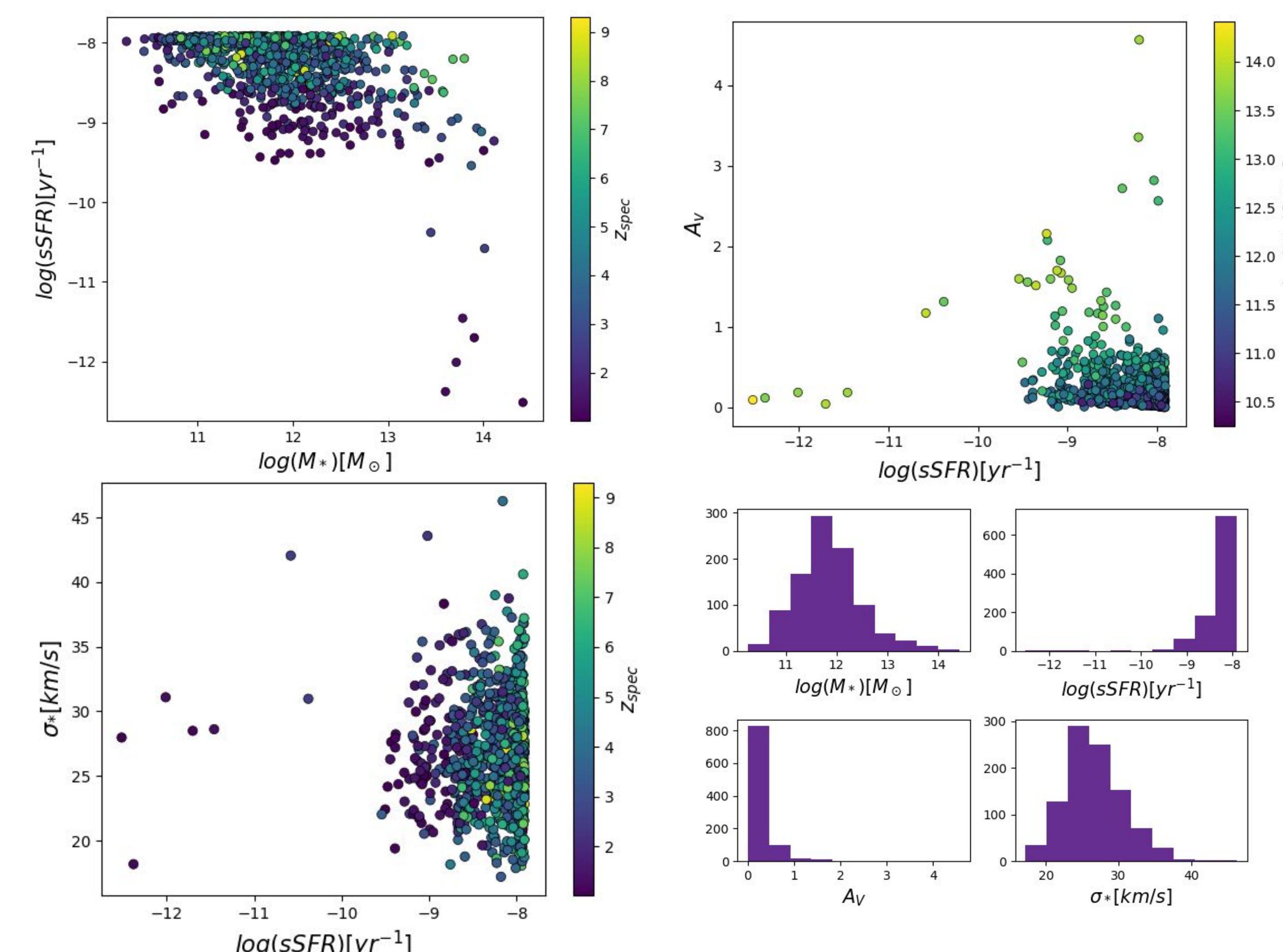


Figure 3. Galaxy properties returned from BAGPIPES SED fitting. a) sSFR as a function of stellar mass. Points are colored based on redshift. Lower redshift sources tend to have lower sSFR. b) A_V versus sSFR colored by redshift. We find no correlation between velocity dispersion and the other properties. c) Stellar mass, sSFR, A_V , and velocity dispersion distributions of the quiescent galaxy sample.

Conclusion

We have identified 958 massive quiescent galaxies from $1 < z < 9$. Our analysis of the galaxy properties reveals that this sample is relatively dust-free and has low stellar velocity dispersion. For our average $A_V = 0.15$, dust is considered to be negligible; however, the few sources with high A_V are the most massive with low sSFR, so it will be necessary to do a further assessment to ensure that these are not dusty impostors. The low velocity dispersion of the sample indicates that these galaxies may not contain AGN outflows.

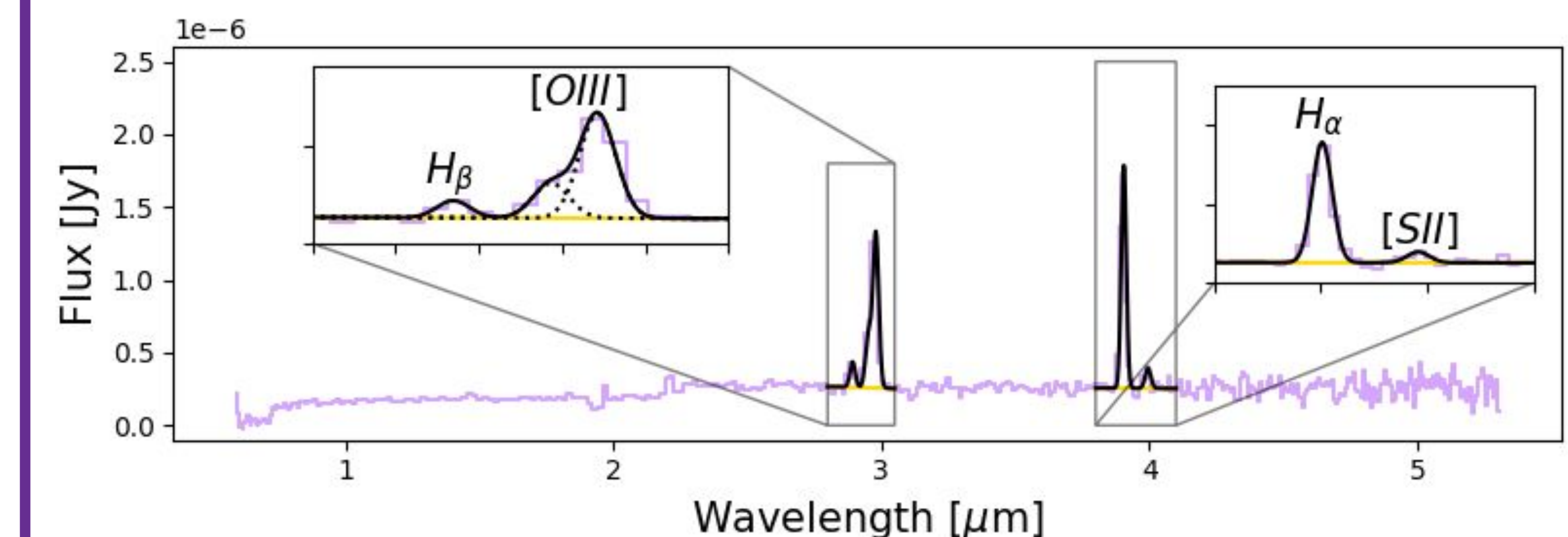


Figure 4. Spectrum of CEERS galaxy at $z \sim 4.5$. Main emission lines are fit using gaussian profiles, and zoomed-in boxes are of the fits. This figure is an example of the line-fitting we will do for the entire sample to identify NLAGN.

With this sample of massive quiescent galaxies, we will **identify NLAGN** through **emission line diagnostic diagrams**. Figure 4 demonstrates the gauss fitting that is necessary to find the amplitude of the emission lines. By comparing emission line ratios, we will investigate how the presence of an NLAGN corresponds with the host galaxies' properties. By studying the **quenching timescales** and calculating the velocity dispersion from the spectroscopic data, we will characterize the AGN feedback of this sample.

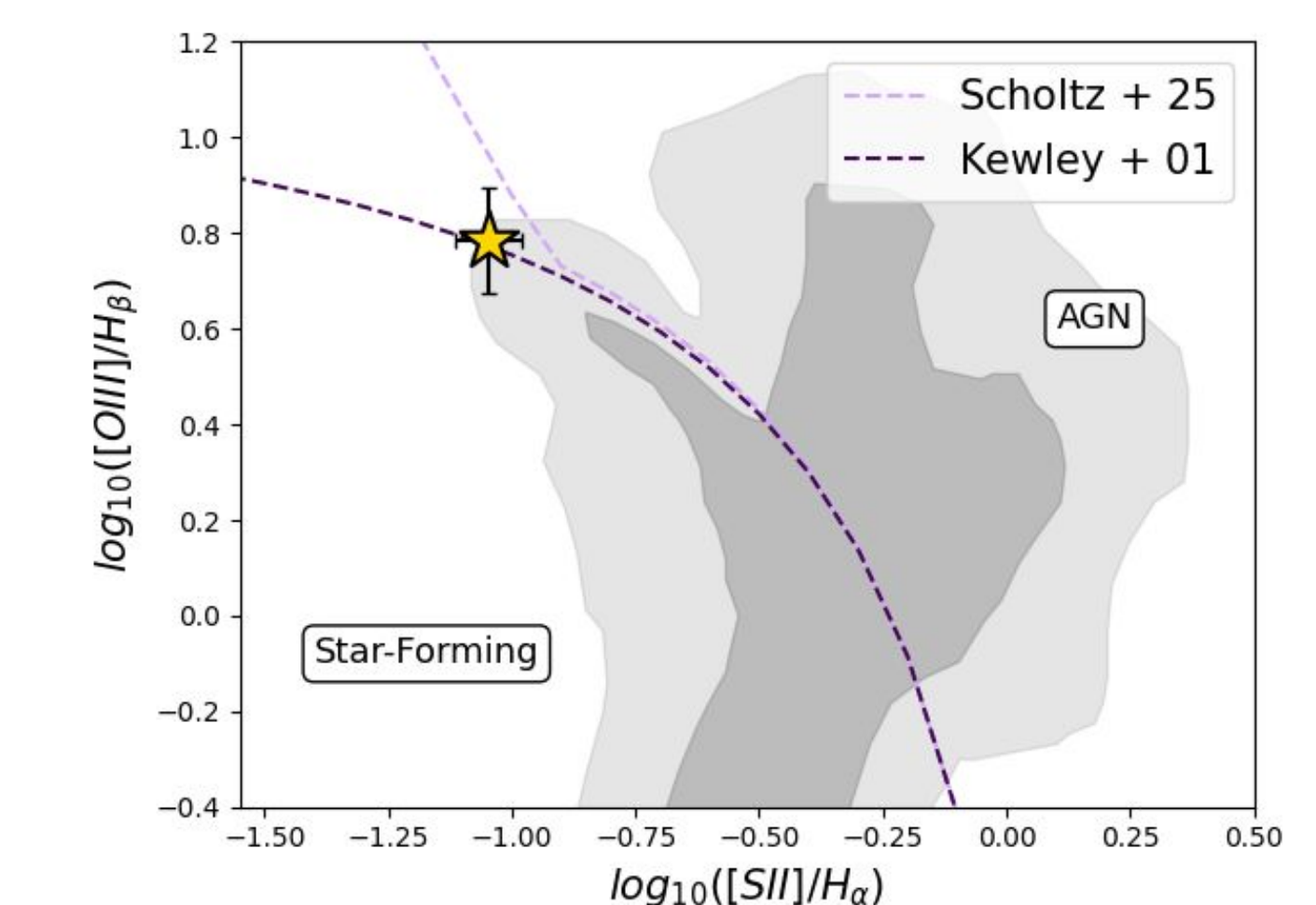


Figure 5. V807 diagram. Dashed lines indicate the demarcation lines from Scholtz et al 2025 [2] and Kewley et al 2001 [7]. Contours are from local galaxies. This source is not classified as an AGN with updated demarcation line.

Acknowledgments

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References

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